

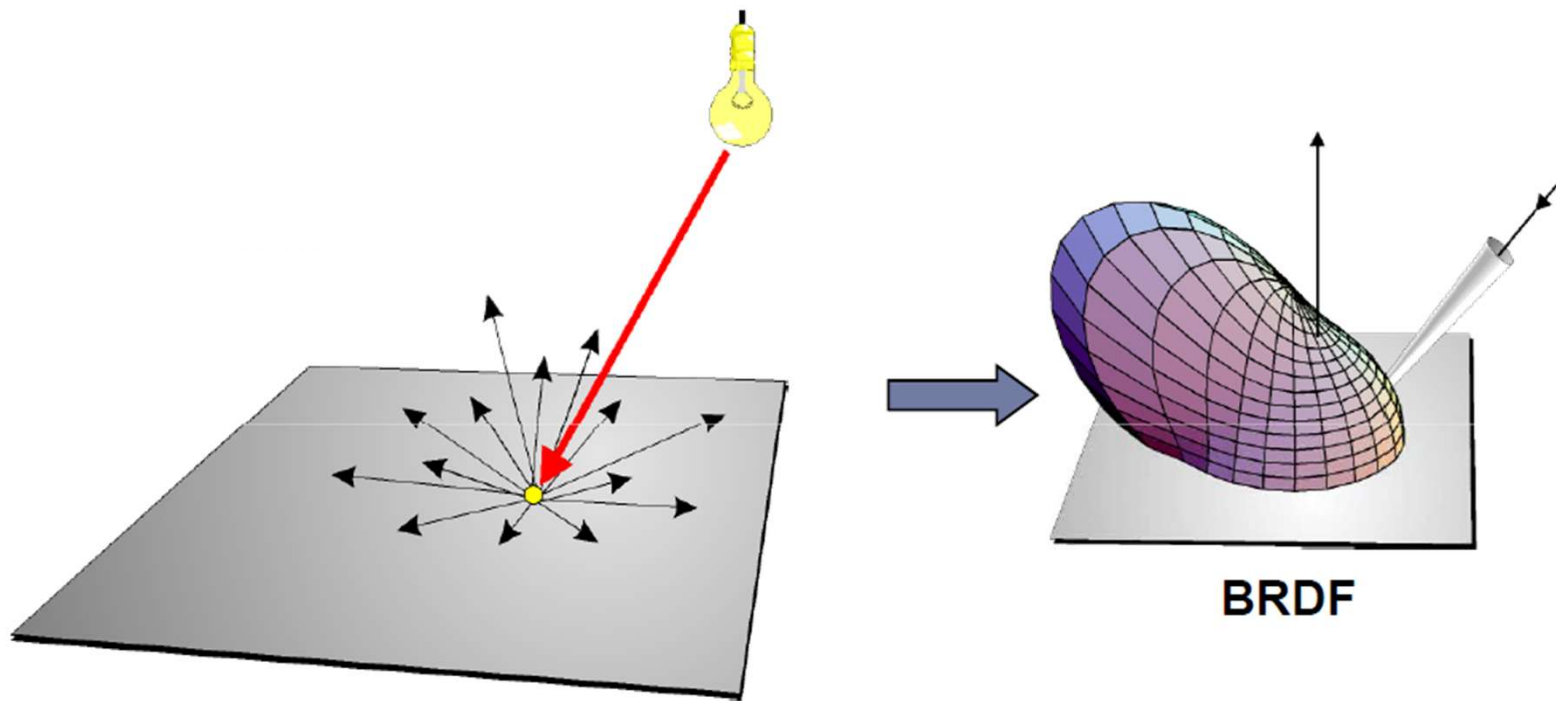
Local Illumination Models

CS7GV3 – Real-time Rendering

Overview

- General definition of Illumination Models
- Local Illumination Models on the Shader
- How do we simulate different light reflection behaviour say between shiny plastic and metal that is the same colour

Scattering



Energy is scattered from a surface in a distribution that depends on the surface's microscopic geometry. This distribution can be described as a function that records the reflected energy per direction = *BRDF*

BRDF

- General light reflected from a point on a surface is categorised by the Bi-directional Reflectance Distribution Function
 - Function of some direction (in local-illumination we are interested in the viewer) and incoming light direction
 - Can be written as

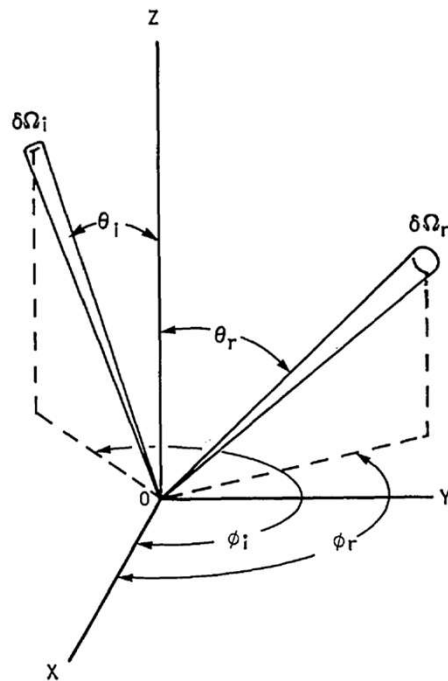


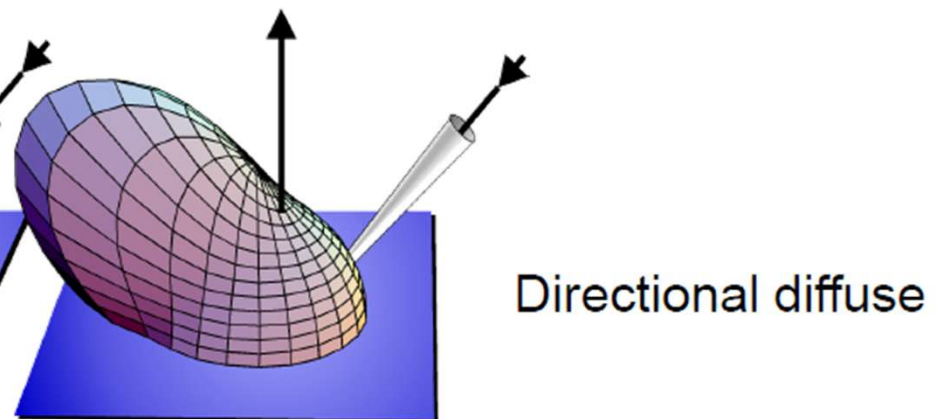
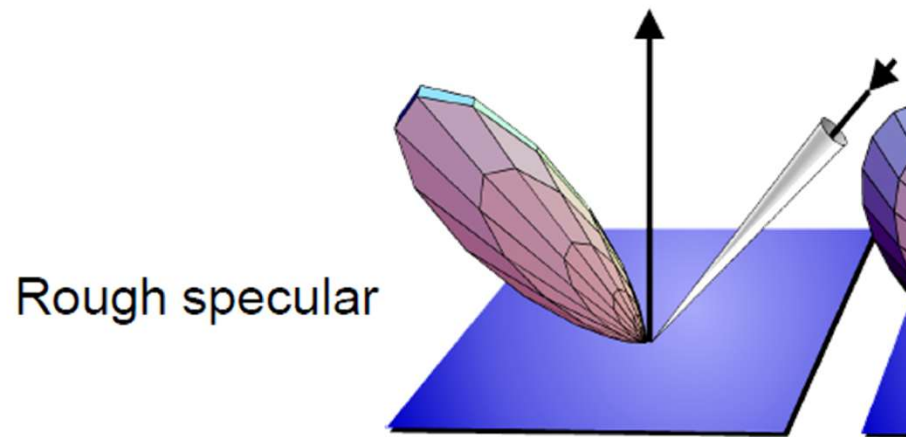
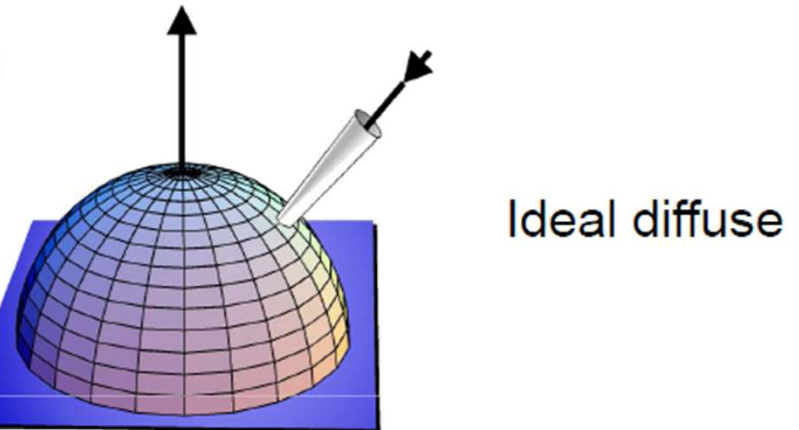
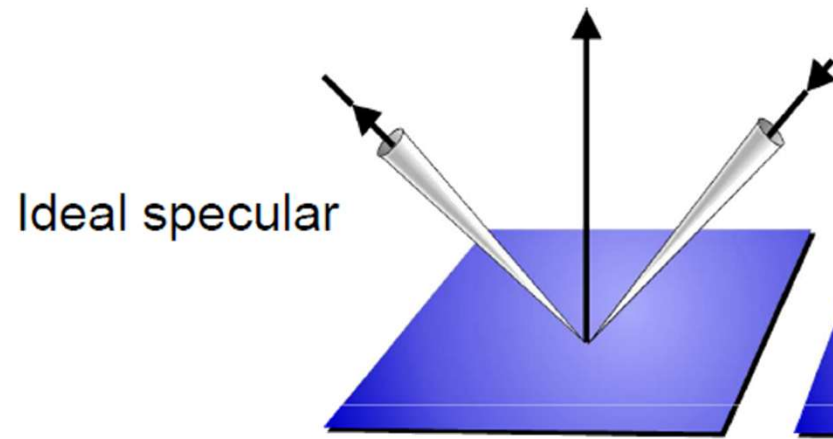
Fig. 1

$$BRDF = f(\theta_{in}, \phi_{in}, \theta_{ref}, \phi_{ref}) = f(\mathbf{L}, \mathbf{V})$$

Citation

Fred E. Nicodemus, "Directional Reflectance and Emissivity of an Opaque Surface," Appl. Opt. **4**, 767-775 (1965);
<https://www.osapublishing.org/ao/abstract.cfm?uri=ao-4-7-767>

BRDF Approximations



Reflectance Equation

- The *reflectance equation relates reflected* radiance to incoming radiance that is scattered according to the surface's BRDF

$$\underbrace{L_r(x, \omega_r)}_{\text{Reflected radiance}} = \int_{\Omega} \underbrace{f_r(x, \omega_i, \omega_r)}_{\text{BRDF}} \underbrace{L_i(x, \omega_i)}_{\text{Incident radiance}} \underbrace{\cos \theta_i d\omega_i}_{\text{Projected Solid Angle}}$$

$\Rightarrow$ **recursive**

Ω = domain of integration

Hemisphere if surface is opaque, otherwise the sphere

Reflectance Equation

- Actually more complex than this:
 - Integrate over entire sphere (for transparency)
 - Need to integrate over time (for exposure, motion blur)
 - Integrate over surface of lens (for aperture effects)
 - Integrate over the visible spectrum (for dispersion etc.)
 - And more...
- More accurate equation:

$$L_r(x, \lambda, \omega_r) = \int_T \int_\Lambda \int_\Omega f_r(x, t, \lambda, \omega_i, \omega_r) L_i(x, t, \lambda, \omega_i) \cos \theta_i d\omega_i d\lambda dt$$

- ...and this doesn't include the camera...
- We'll need to simplify for real-time

Rendering

- Rendering: determine the *most appropriate colour* to assign to a pixel associated with an object in a scene.
- The visible colour of an object at a point depends on:
 - geometry of the object at that point (*normal direction*)
 - position, geometry and colour of the light sources (*luminaires*)
 - position and visual response of the viewer / camera
 - *surface reflectance properties of the object at that point*
 - scattering by any *participating media* (e.g. *smoke, rising hot air*)
- Units:
 - Physically based: radiance, irradiance, power, energy etc.
 - Many talk about intensity which is non-physically based
- Most pipelines ignore physics units: everything is a tweakable from 0.0 to 1.0

Local Illumination Requirements

- Illumination Model (or Shading Model)

- Approximation of a BRDF

$$L_r(x, \omega_r) = \int_{\Omega} \boxed{f_r(x, \omega_i, \omega_r)} L_i(x, \omega_i) \cos \theta_i d\omega_i$$

- Surface Model

- Defines variation in BRDF over surface (e.g. texture)

$$L_r(x, \omega_r) = \int_{\Omega} \boxed{f_r(x, \omega_i, \omega_r)} L_i(x, \omega_i) \cos \theta_i d\omega_i$$

- Lighting Source Model

- Defines the source of incoming illumination

$$L_r(x, \omega_r) = \int_{\Omega} f_r(x, \omega_i, \omega_r) \boxed{L_i(x, \omega_i)} \cos \theta_i d\omega_i$$

Light Sources

- In Real-time rendering we normally employ *isotropic point light sources defined by position, colour and power*
 - Isotropic $\Rightarrow$ radiates energy equally in all directions
 - Point $\Rightarrow$ has no area!
- The power per unit area (irradiance E) at a point x at distance r from the point light source may now be determined:
 - power is radiated isotropically, through the sphere centered at the light position
 - at a distance r from the source, the surface area of this sphere is $4\pi r^2$
 - therefore the power per unit area at x is:

$$E = \frac{\Phi}{4\pi r^2}$$

- note: this assumes the surface at x is *perpendicular to the direction to the light source*.

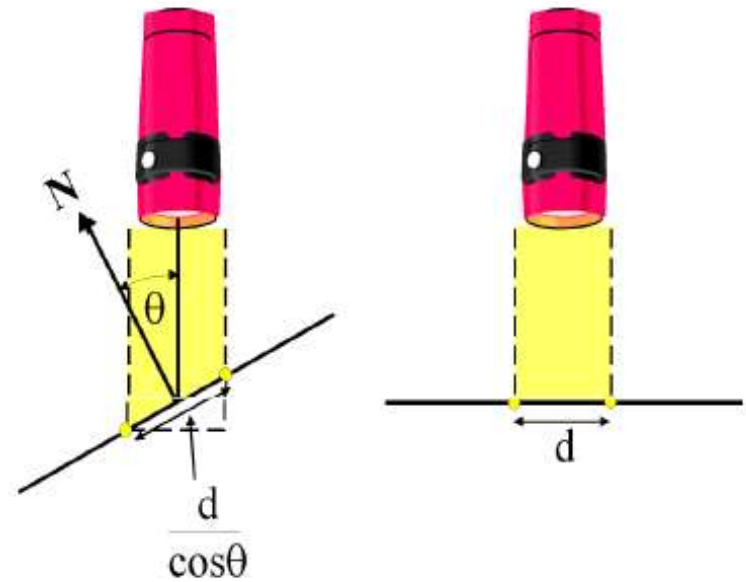
The Cosine Rule

- A surface oriented perpendicular to a light source will receive more energy (and thus appear brighter) than a surface oriented at an angle to the light source.
- The irradiance E is inversely proportional to area:

$$E \propto \frac{1}{A}$$

- As the area increases the irradiance decreases therefore:

$$E = \frac{\Phi \cos \theta}{4\pi r^2}$$



- As θ increases, the irradiance and thus the brightness of a surface decreases by $\cos \theta$

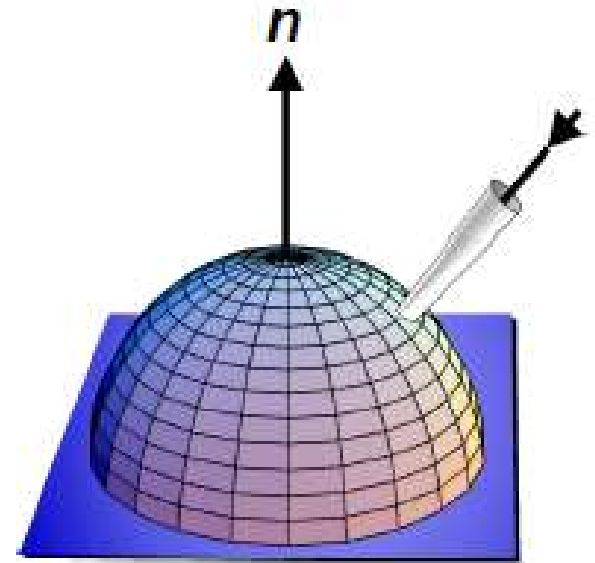
Lambertian Illumination

- We can use the cosine rule to implement shading of *Lambertian or diffuse surfaces*.
- Diffuse surfaces reflect light in all directions equally:
 - BRDF is a constant with respect to reflected direction
 - surface may be characterized by a reflectance ρ_d rather than a BRDF:

$$\rho_d(x) = \frac{\Phi_i(x)}{\Phi_r(x)}$$

- the reflectance gives the ratio of the total reflected power to the total incident power

$$f_r(x) = \frac{\rho_d(x)}{\pi}$$



Lambertian Illumination

- To shade a diffuse surface we need to know:
 - normal to the surface at the point to be shaded
 - diffuse reflectance of the surface
 - positions and powers of the light source in the scene
- Contribution from a single source is given by:

$$L_{r,d}(\mathbf{x}, \cdot) = \frac{\rho_d}{\pi} \cos \theta \frac{\Phi_L}{4\pi r^2}$$

No dependence on directions

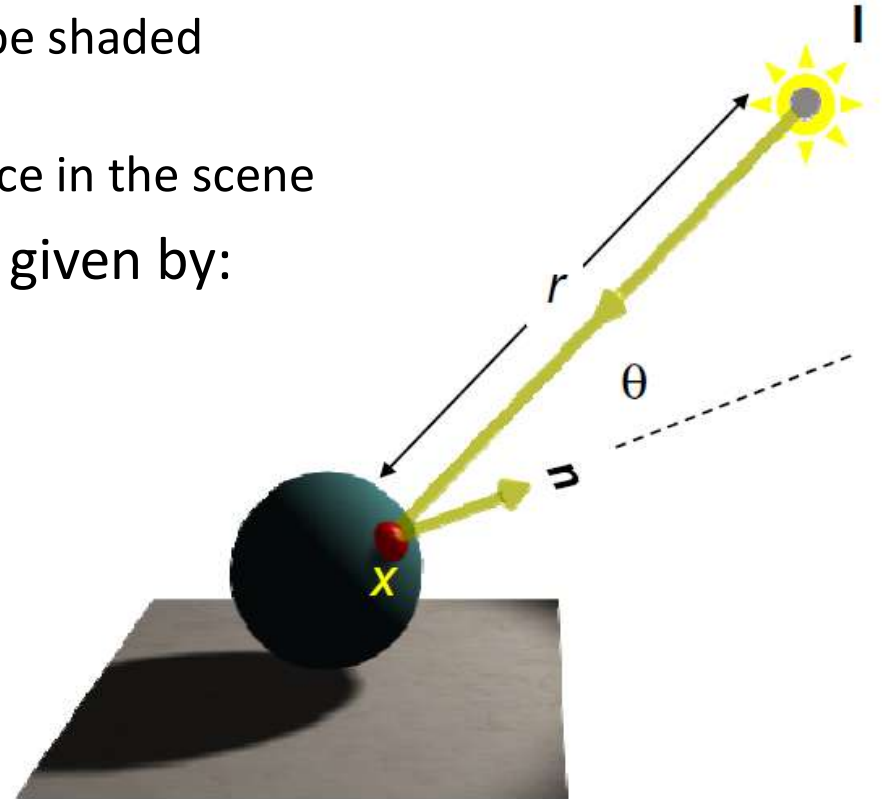
$$= \frac{\rho_d}{\pi} (\mathbf{n} \cdot \mathbf{l}) E$$

In practice replaced by single co-efficient K_d

- This is *local illumination*

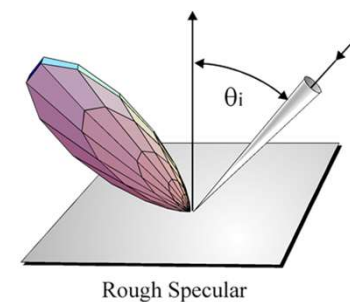
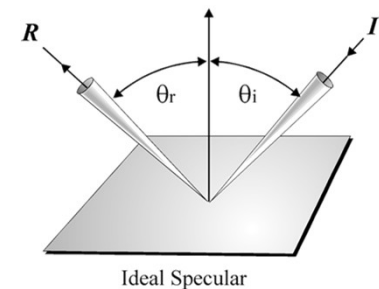
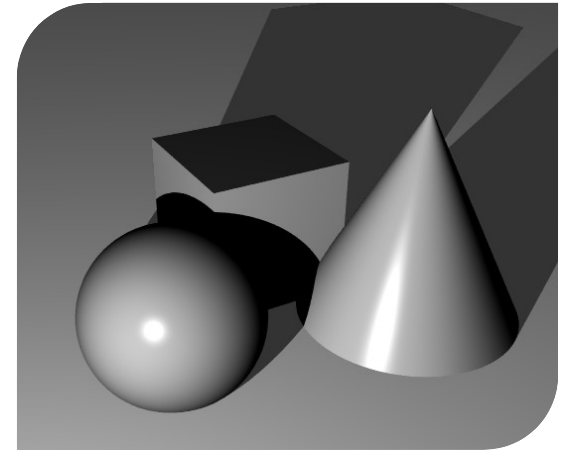
In practice: replaced by the light intensity L
(attenuation accounted for elsewhere)

$$I_d = k_d (\mathbf{n} \cdot \mathbf{l}) L_d$$

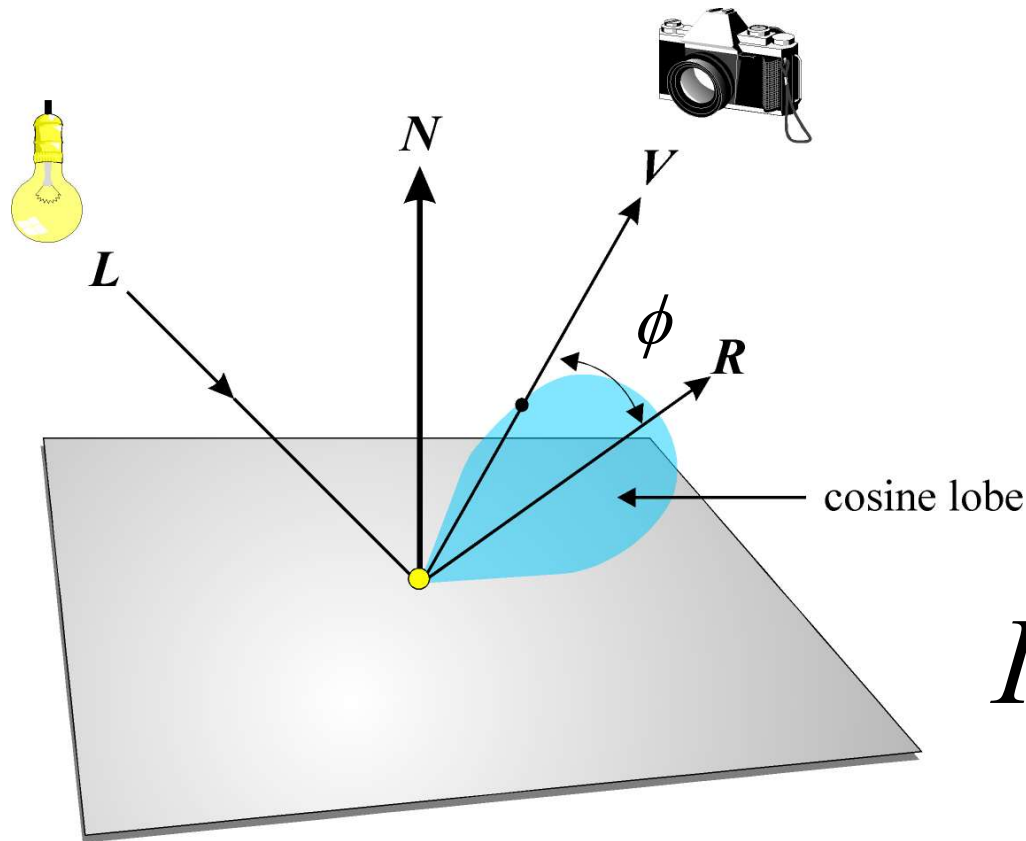


Specular Highlights

- Specular surfaces exhibit a high degree of coherence in their reflectance, i.e. the reflected radiance depends very heavily on the outgoing direction.
- An ideal specular surface is optically smooth (smooth even at resolutions comparable to the wavelength of light).
- Most specular surfaces (rough specular) reflect energy in a tight distribution (or lobe) centered on the optical reflection direction:



Phong Model of Specular Reflection



- Intensity reflected light depends on:
 - Viewer direction
 - Incoming light direction
 - Light colour + brightness (L)
 - (α) Phong exponent/shininess
 - (k_s) specular reflectivity coefficient

$$I_S = k_S L_S \cos^\alpha \phi$$

Local model: only consider reflections of light sources. Assume that the BRDF of shiny surfaces may be approximated by a spherical cosine function raised to a power (known as the Phong exponent, indicative of the shininess).

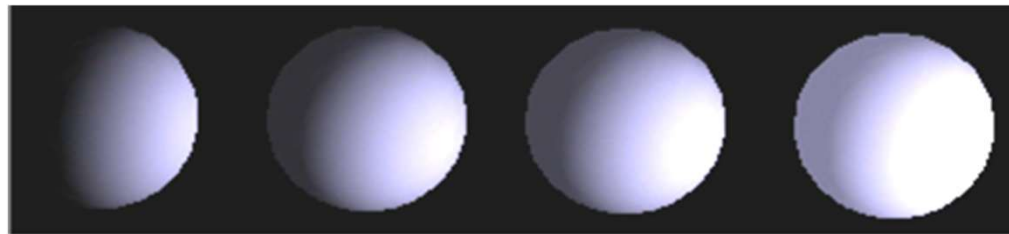
Ambient Light

- Objects lit by ambient light are lit evenly on all surfaces in all directions
- Certain lights e.g. tube lights in class-rooms or kitchens try to achieve this by using large diffusers
- Ambient illumination is characterised by an intensity L_a identical at every point in the scene



$$I = L_a k_a$$

where k_a is proportion of ambient light reflected

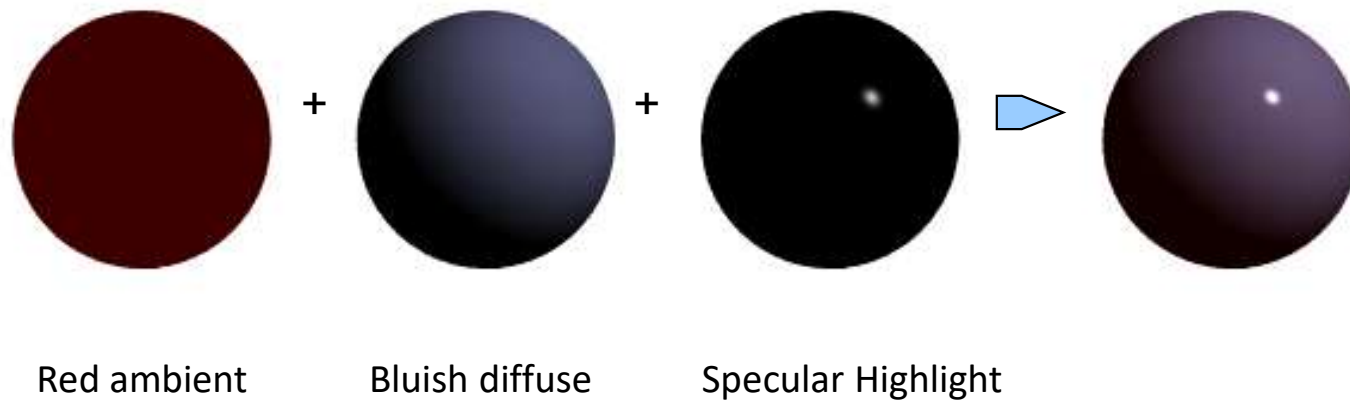


$$I = L_a k_a + I_d$$

N.B. Usually we add only very small amounts of ambient light

Putting it all together

- The intensity of light from one point is a sum of the diffuse, specular and ambient components:



Phong Illumination Model

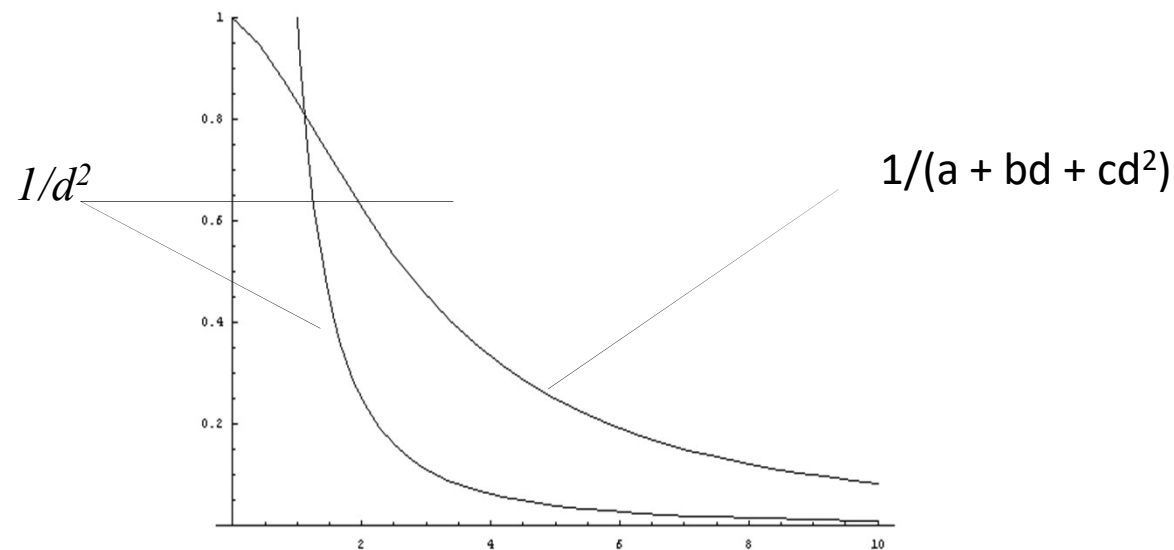
```
//.vs
varying vec3 lightDir, normal, pos;
void main()
{
    lightDir = normalize(vec3(gl_LightSource[0].position - gl_ModelViewMatrix * gl_Vertex));
    pos = vec3(gl_ModelViewMatrix * gl_Vertex);
    normal = normalize(gl_NormalMatrix * gl_Normal);
    gl_Position = gl_ModelViewProjectionMatrix * gl_Vertex;
}
```

Phong Illumination Model

```
//.fs
varying vec3 lightDir, normal, pos;
void main()
{
    float i_diffuse, i_specular, i_ambient;
    vec3 n;
    vec3 viewDir = normalize(-pos);
    vec3 refDir = normalize(-reflect(lightDir, normal));
    vec4 color;
    n = normalize(normal);
    i_diffuse = max(dot(lightDir,n), 0.0);
    i_specular = max( pow( dot(refDir, viewDir), 100), 0.0);
    i_ambient = 1;
    color = vec4(0.5, 0.5, 1.0, 1) * i_diffuse + vec4(1, 1, 1, 1)* i_specular + i_ambient*vec4(.2, 0, 0, 1);
    gl_FragColor = color;
}
```

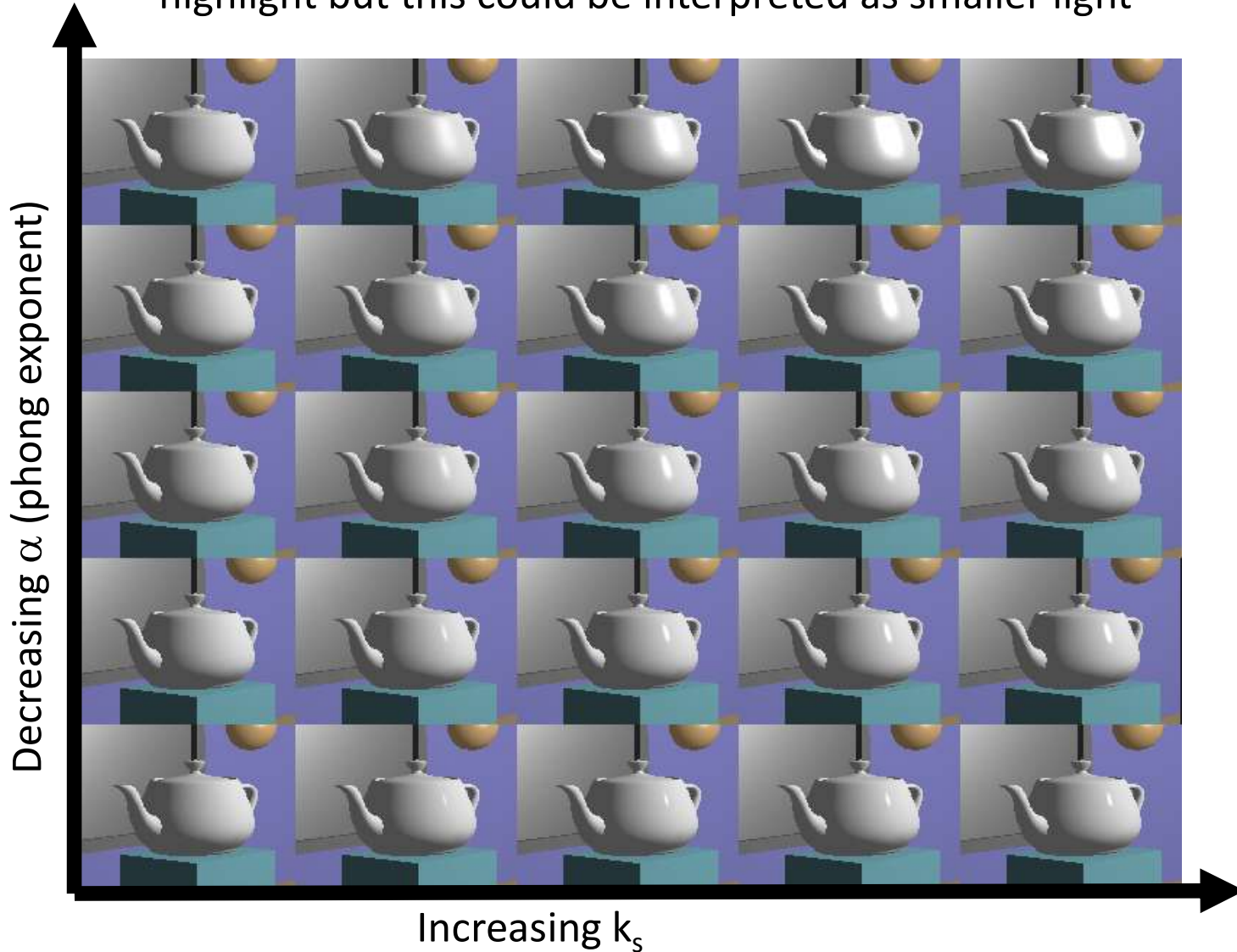
Light Attenuation

- Physically accurate attenuation
 - Attenuation factor = $1/d^2$
 - d is distance from light to surface
 - In practice this is very harsh – falls off too quickly
- Quadratic Attenuation with softer fall off
 - Attenuation factor = $1/(a + bd + cd^2)$



Phong Shading Limitations

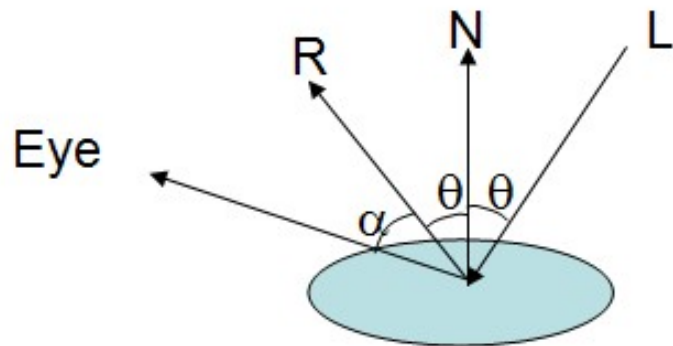
For a point light source: increasing shininess leads to smaller highlight but this could be interpreted as smaller light



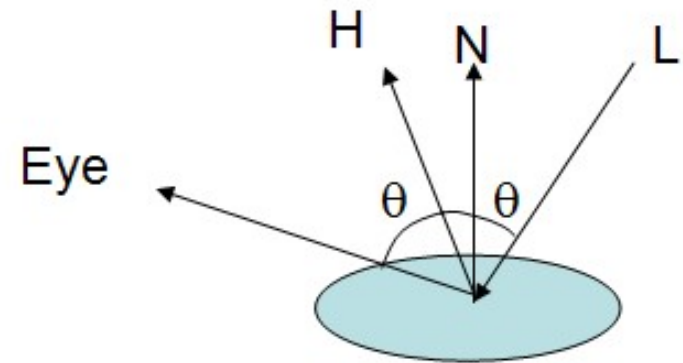
Phong & Blinn-phong models (1/2)

[1975]

- The formula for specular is the Phong model
 - not physically correct
 - looks nice in practice and very simple to evaluate
- Blinn simplification
 - use angle with half-vector
 - also standard in Computer Graphics



Phong

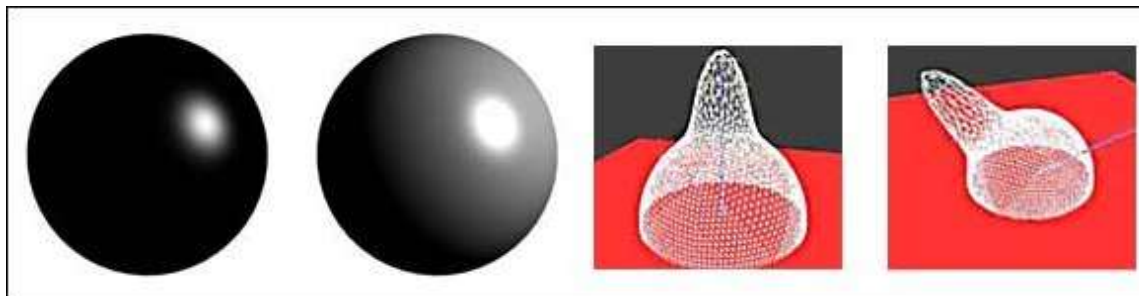


Blinn-Phong

Phong & Blinn-phong models (2/2)

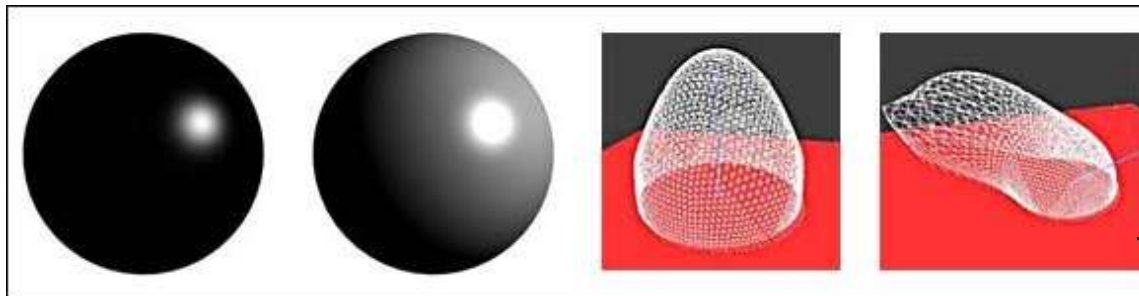
- Difference is a matter of taste!
- Blinn-phong tends to be more predictable

Phong model

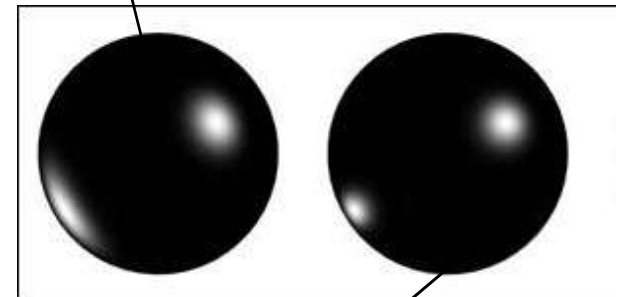


specular

specular+diffuse



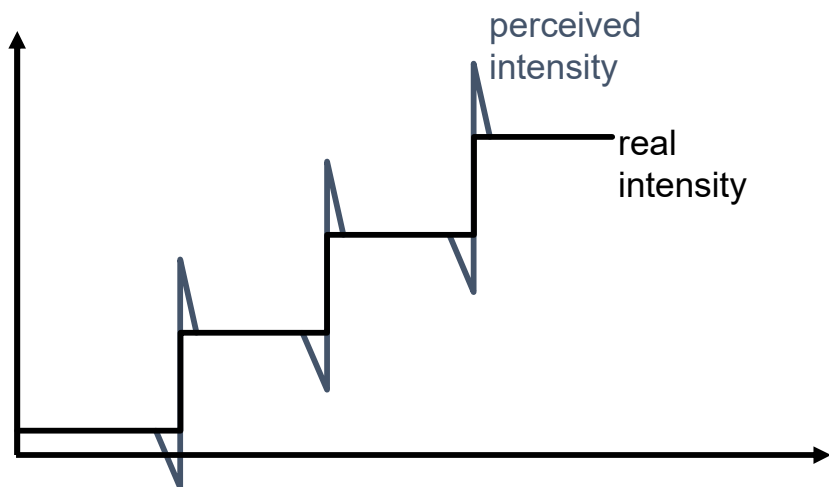
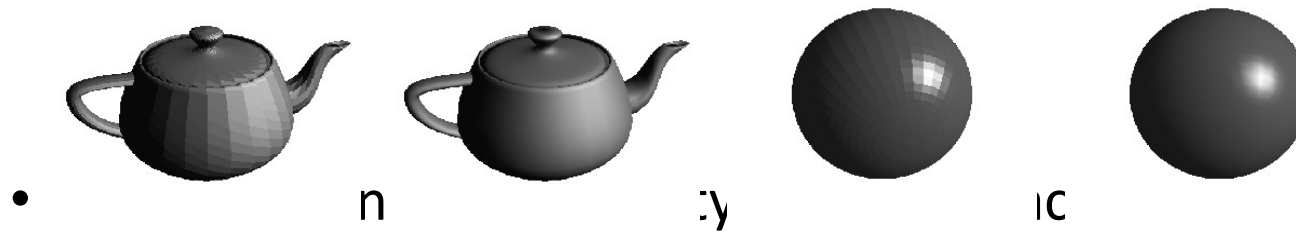
Blinn-Phong model

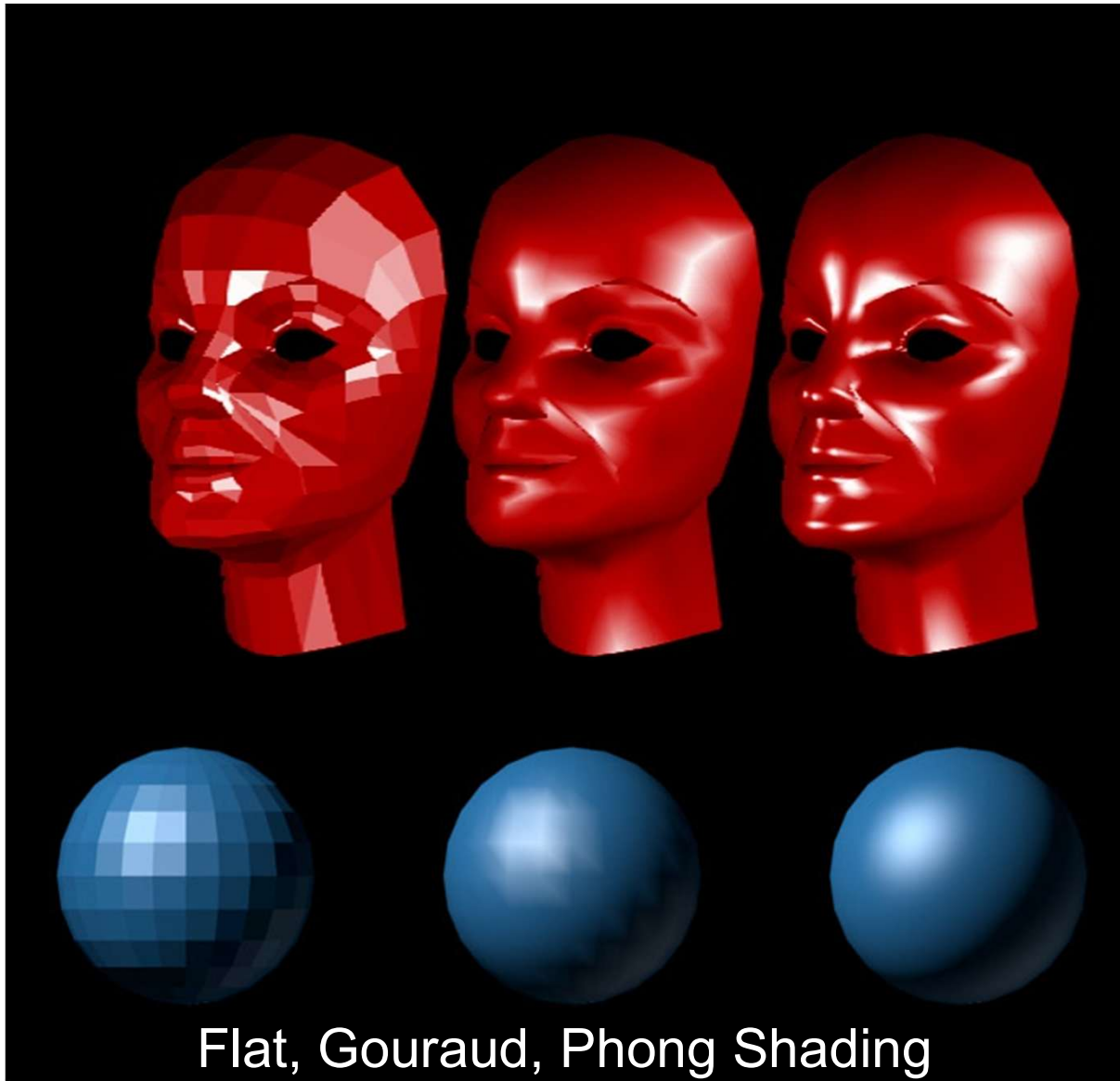


Shading In-Brief

Flat vs. Gouraud shading

- Flat shading creates “faceted” objects
 - requires highly tessellated surfaces





Gouraud vs. Phong shading

- Means per-pixel vs. per vertex shading
- Per pixel is much nicer
 - And more “correct”



A historical note on graphics terms...

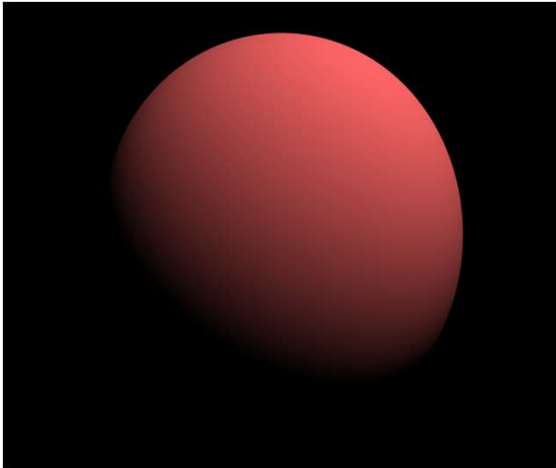
- Lambertian Illumination Model
 - Uses diffuse only reflectance
- Phong Illumination Model
 - Simulates diffuse, specular and ambient reflectance terms
- Gouraud Shading
 - Evaluate lighting model at vertices only, interpolate colour for each pixel
- Phong Shading
 - Evaluate Phong Model at each pixel
- These distinctions are less relevant with PS and VS shaders

Advanced Illumination

Phong Illumination

- Local illumination is mainly combination of
 - Diffuse
 - Perfect Diffuse when light is scattered equally in all directions
 - E.g. from perfect matte surface
 - Specular
 - Perfect specular when light strikes mirror-like surface
 - Thin beam of light in reflected direction $\theta_i = \theta_r$
 - Combining elements of both approximates some complexities of real-behaviour
- ... However Phong itself misses a few things due to its specific approximations

Real-World BRDFs



Complex Surface Scattering Examples



Multi-layered Surfaces



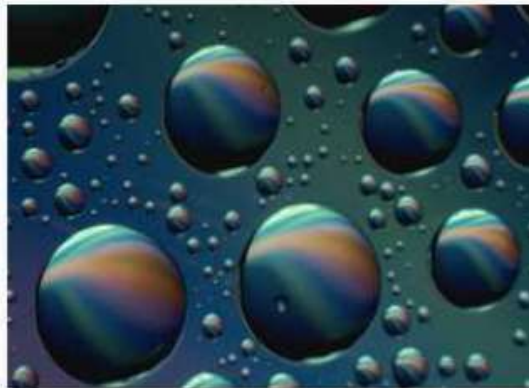
Coloured Glass



Human Skin



Stone



Thin Film



Tarnished Metal

Lighting Models: Beyond Phong

- Lighting models

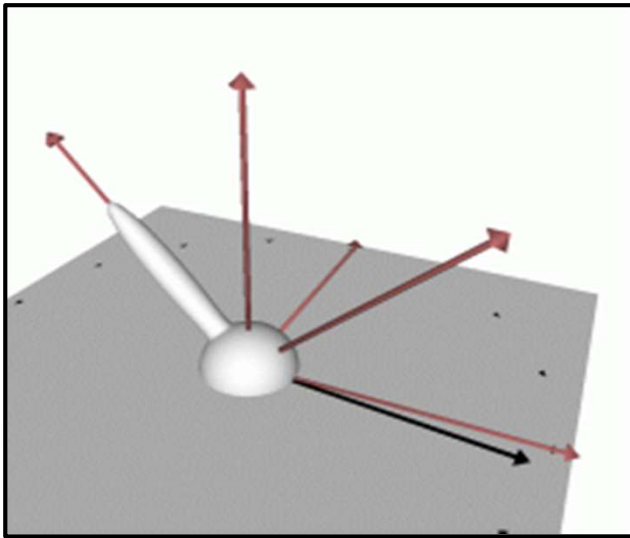
- Torrance-Sparrow [1967]
- Blinn [1977]
- Cook-Torrance [1982]
- Ward Anisotropic [1992]
- Oren-Nayar [1994]
- Schlick [1994]
- He-Torrance-Sillion-Greenberg [1991]
- Lafortune et al. [1997]
- Ashikhmin-Shirley [2000]
- Hapke/Lommel-Seeliger [1963]
- Strauss [1990]
- Poulin-Fournier [1990]
- Minnaert [1941]



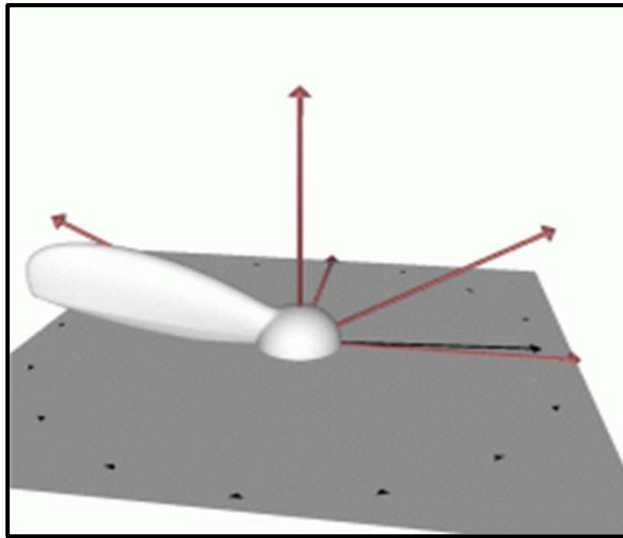
Minnaert - An empirical model with applications in Fabric modelling (velvet, satin)

BRDF

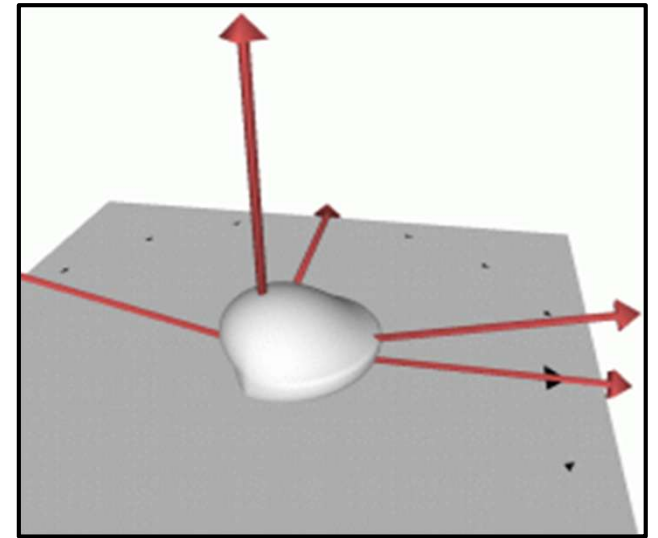
- The bidirectional reflectance distribution function is surface



Phong BRDF



Cook-Torrance BRDF



Oren-Nayar BRDF

- Given incoming light direction, v_i , and outgoing light direction, v_o , how much light is reflected?
- Can depend on wavelength, λ .

Measuring BRDFs

- Gonioreflectometer: measure reflected light for various light and reflection directions of the hemisphere.



Real BRDF



Selection of Models described in Watt Polcarpo – the Computer Image

Iron



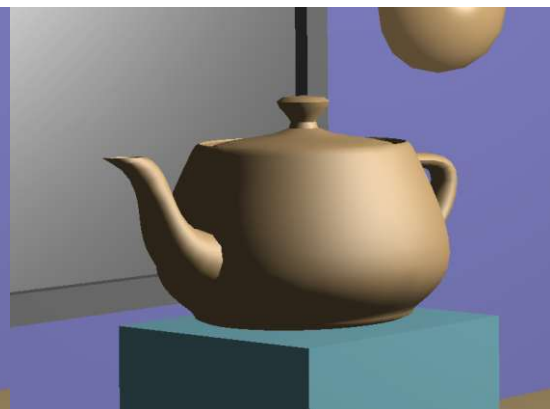
Steel



Stainless Steel



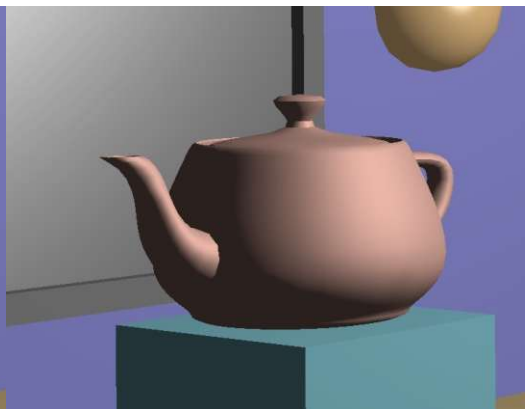
Antique Brass



Polished Brass



Copper



Bronze



Lead



Polished Gold



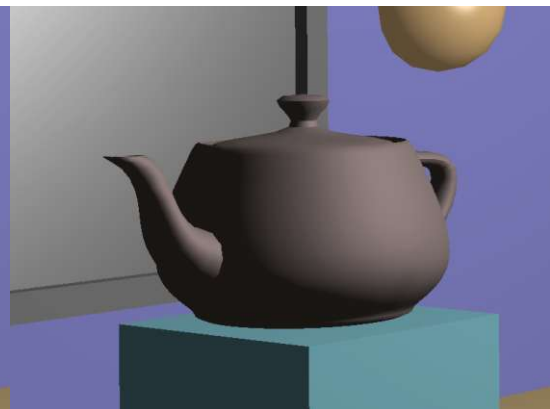
Silver Plate



Chromium Plate



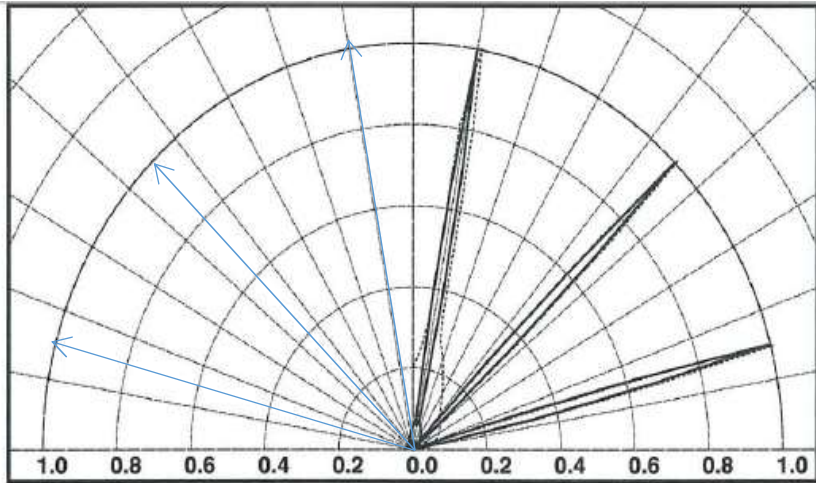
Graphite



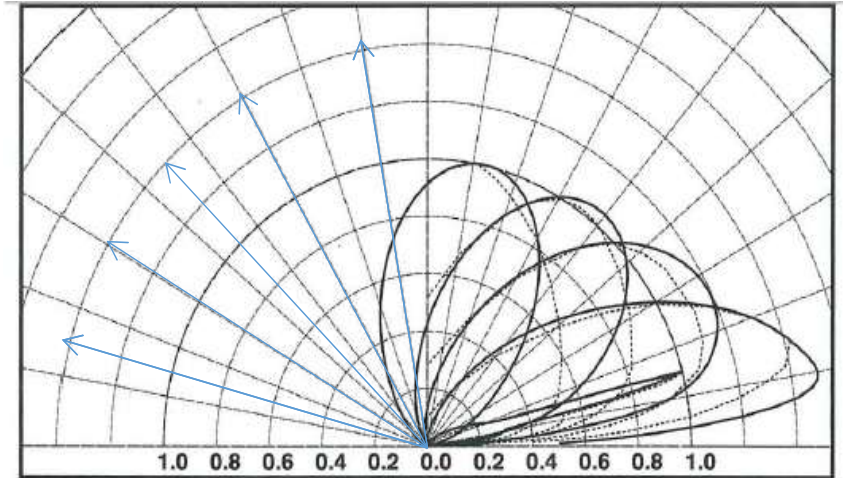
Real-BRDF

- In reality
 - function is not monochromatic: different behaviours for different wavelengths (separate BRDF required)
 - Scattering of light by atmosphere
 - Sub surface scattering
- Formally: BRDF is a four-dimensional function that defines how light is reflected at an opaque surface.
 - Based on (with respect to the surface normal n) [Nicodemus 1965]
 - incoming light direction, ω_i ,
 - outgoing direction, ω_o ,
 - return the ratio of reflected radiance exiting along ω_o to the irradiance incident on the surface from direction ω_i .
 - has units sr^{-1} , with steradians (sr) being a unit of solid angle.

Real BRDF – variant with λ



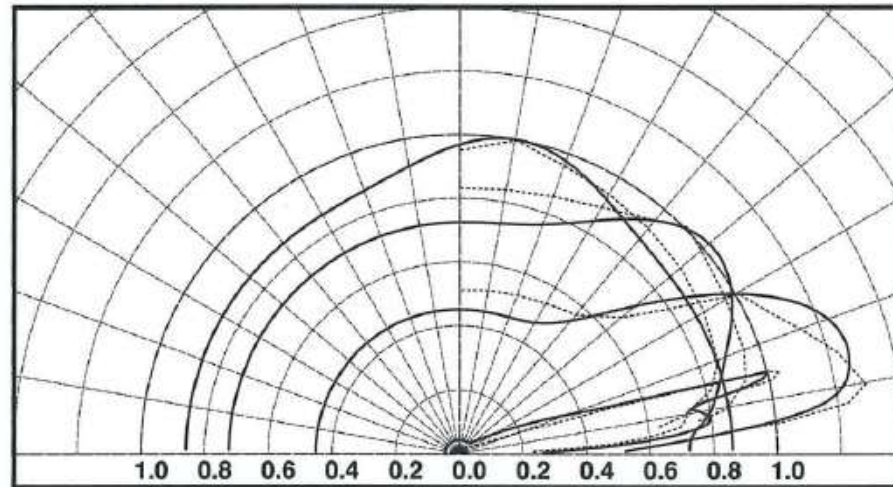
- Normalized BRDF's of roughened aluminum at
 - incidence angles of $\theta_i = 10^\circ, 45^\circ$, and 75° .
 - wavelength $\lambda = 2.0 \mu\text{m}$.
- Strong specular reflection at this wavelength



- Normalized BRDF's of roughened aluminum at
 - incidence angles of $\theta_i = 10^\circ, 30^\circ, 45^\circ, 60^\circ$, and 75° .
 - $\lambda = 0.5 \mu\text{m}$.
- Strong directional diffuse and emerging specular reflection at this wavelength.

- Images From: Xiao D. He, Kenneth E. Torrance, Francois X. Sillion, and Donald P. Greenberg. 1991. "A comprehensive physical model for light reflection." In *Proceedings of the SIGGRAPH '91*.

Real BRDF – variant with θ



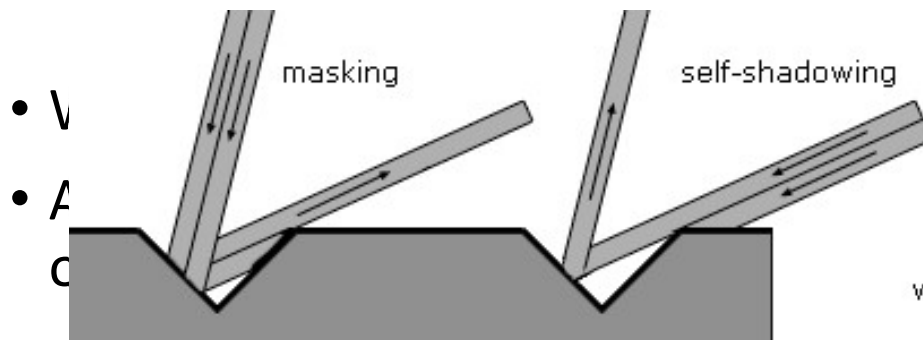
- Normalized BRDF's of roughened magnesium oxide ceramic
 - incidence angles of $\theta_i = 10^\circ, 45^\circ, 60^\circ$ and 75°
 - $\lambda = 0.5 \mu m$.
- Strong uniform diffuse and emerging specular reflection
- Image From: Xiao D. He, Kenneth E. Torrance, Francois X. Sillion, and Donald P. Greenberg. 1991. **"A comprehensive physical model for light reflection."** In *Proceedings of the SIGGRAPH '91*.

Perfect Reflection Models

- Most imitations of real BRDF obtained through small changes to specular component – usually very subtle effects
- We will look at:
 - Empirically spread perfect diffuse. Phong (1977) : already discussed this
 - Physically based specular. Blinn (1977), Cook-Torrance (1982)

Torrance-Sparrow (1967)

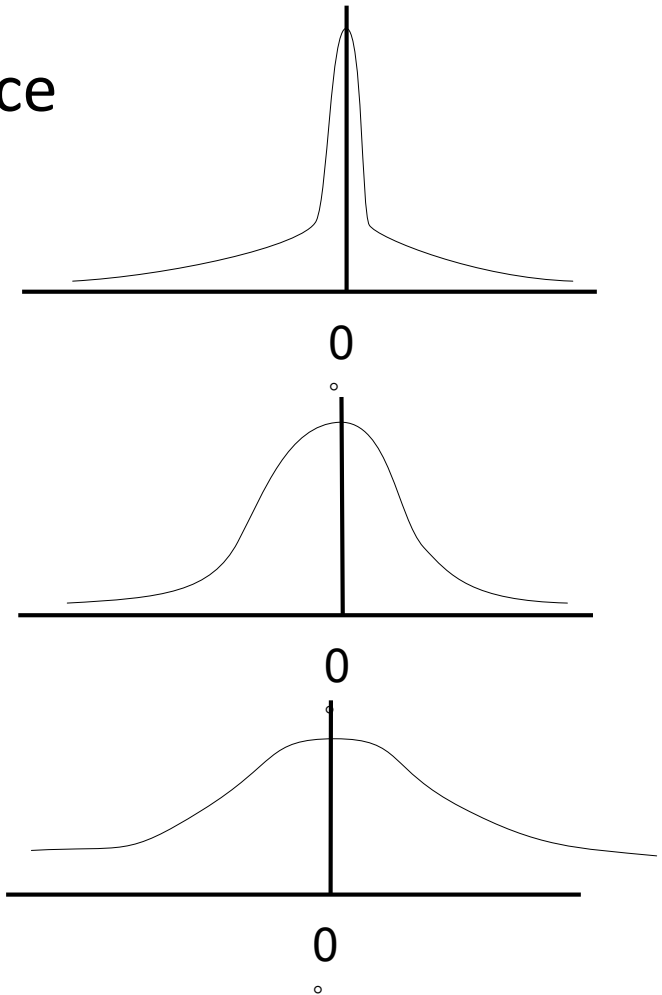
- Gaussian distribution of microfacets:
- Analysis of the geometrical attenuation in symmetrical V-shaped grooves
 - Masking, shadowing



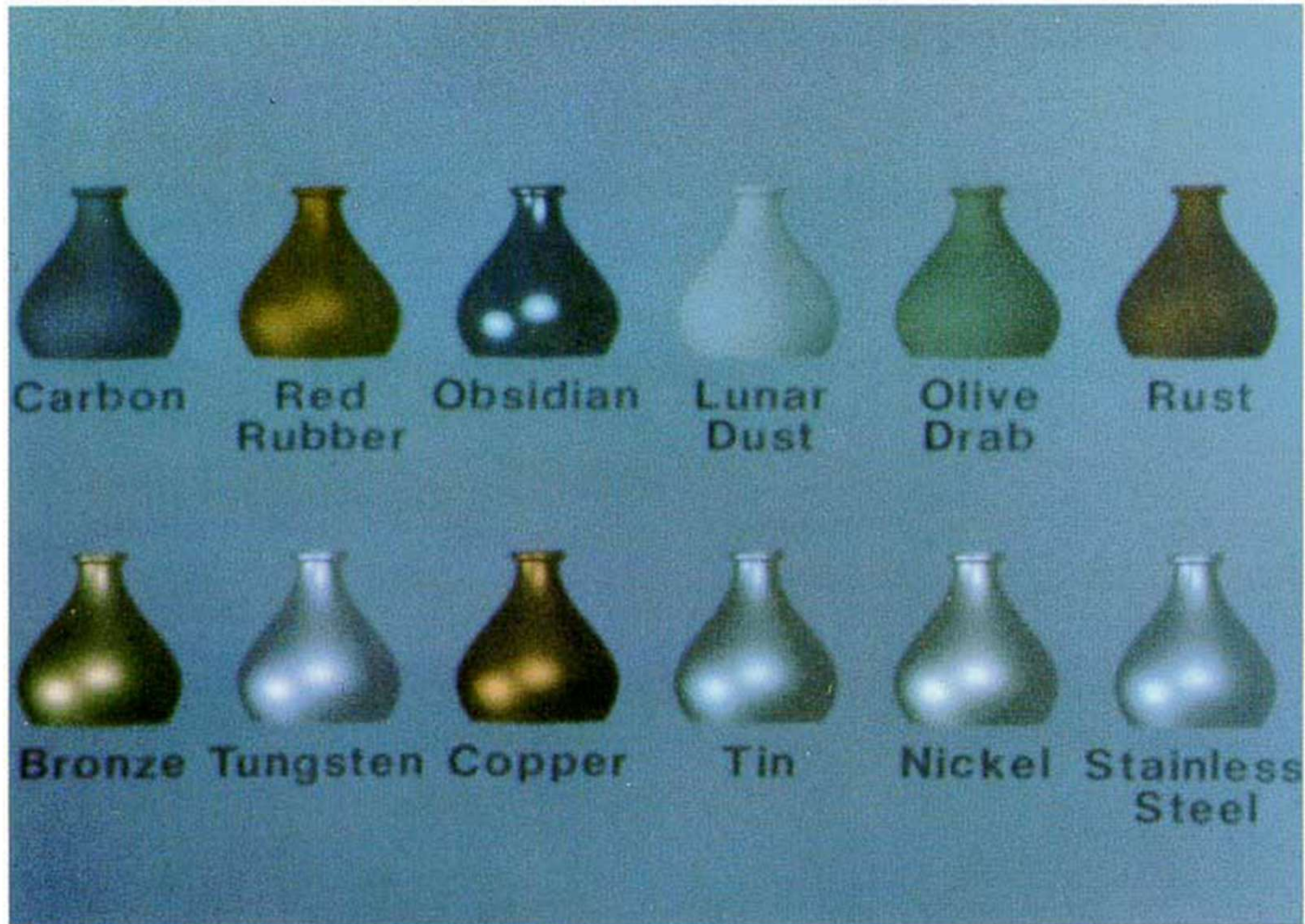
and their lighting models

Distribution of microfacets

- Phong-specular assumes uniform surface
 - In reality : not necessarily uniform



Cook-Torrance results (1982)



Cook-Torrance



☐ Metals

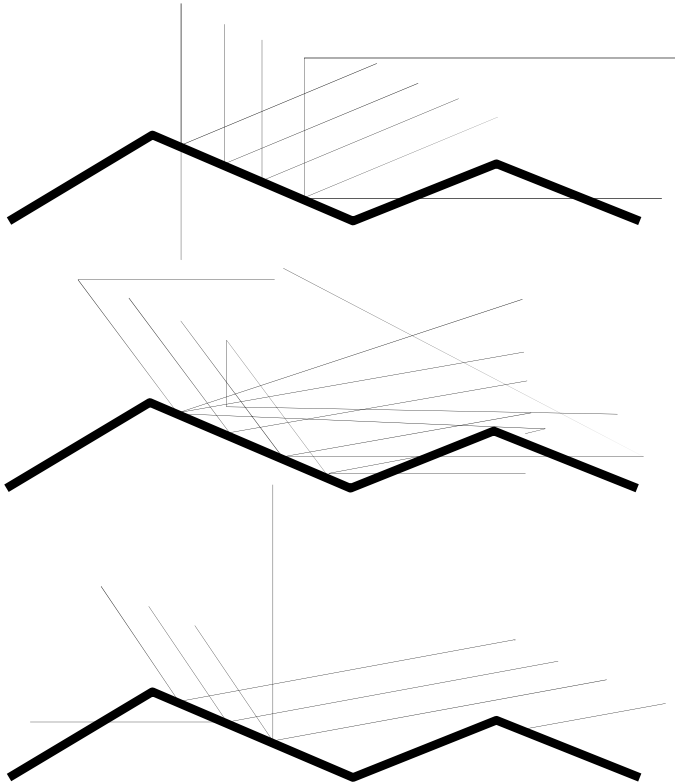
☐ Copper, Tin, Nickel, Stainless Steel

☐ Nonmetals

☐ Carbon, Rubber, Obsidian, Matte finish

Cook-Torrance (1982)

■ G: geometrical attenuation



■ No attenuation: $G = 1$

■ Masking : $G =$

■ Shadowing: $G =$

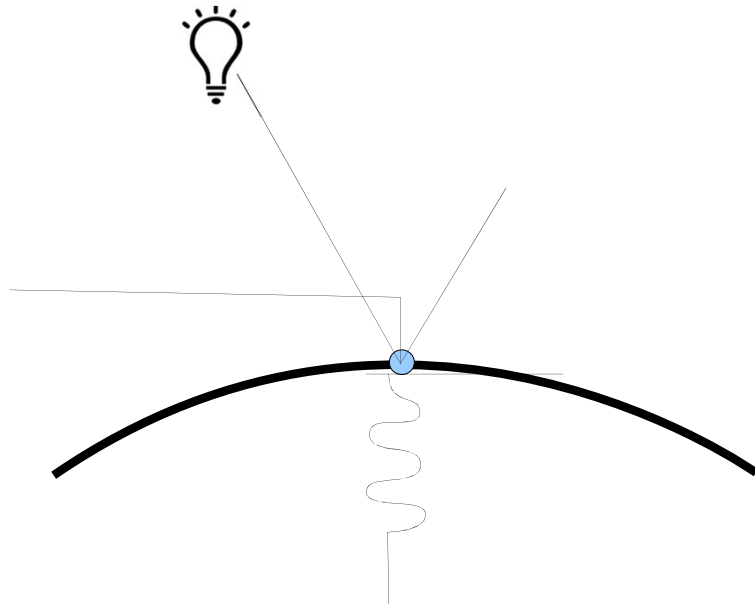
Consider all 3 states by taking the minimum over all possibilities:

Cook-Torrance

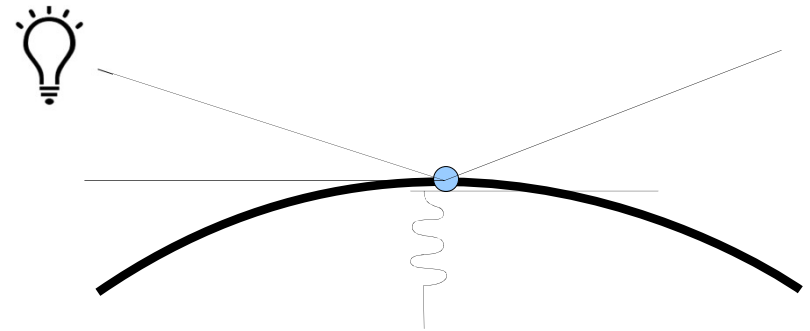
- Blinn (1977) not to be confused with Blinn-Phong shading
 - Microfacets are ellipsoids
 - c is the eccentricity of the ellipsoids
 - $c=0$: very shiny
 - $c=1$: very diffuse
- Cook Torrance extended Blinn to account for spectral composition of highlights with dependency on
 - material type
 - angle of incidence
- Subtle effect on size and colour of the highlight compared to Phong – only changes the specular component
- For shiny metallic surfaces

Fresnel Effect : Imperfect reflection

- Degree of absorption depends on angle of incidence



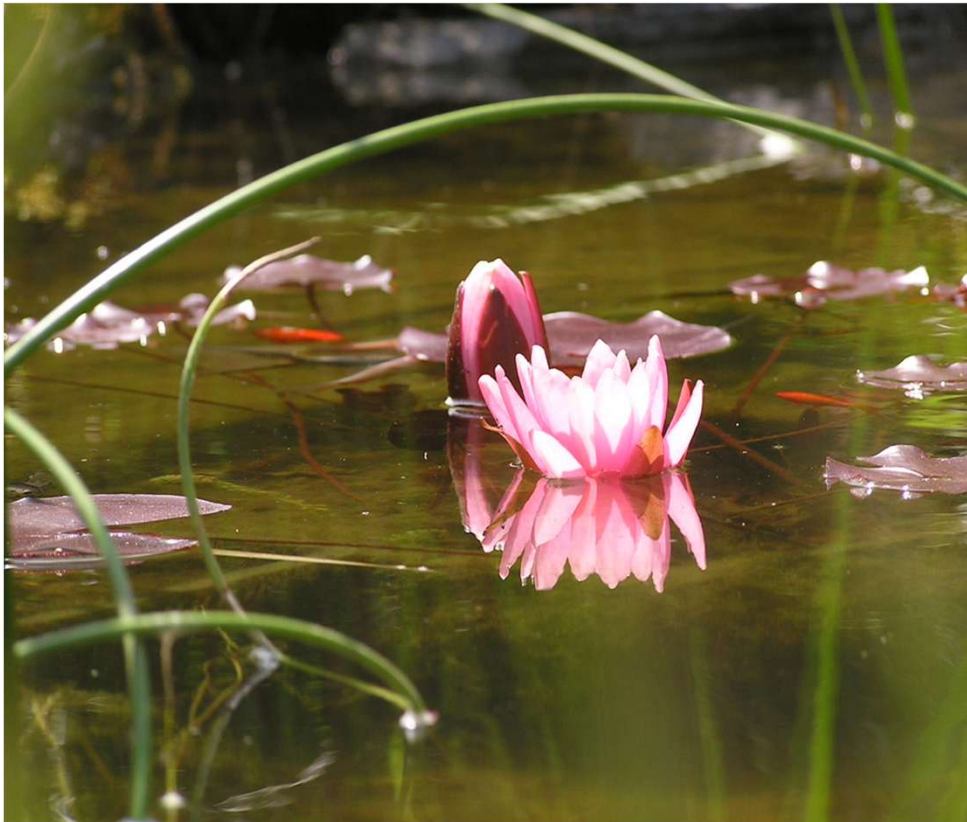
More absorption



Less absorption

Fresnel Effect

- Has major impact on water rendering
- Degree of **refraction** depends on angle of incidence



Ward

- Physically based
 - Measurement on real material
 - Gonio-réfectometer
 - Data sets
- Approximation by gaussians
- Anisotropic model

